DEWDRIVE · FEASIBILITY ASSESSMENT

DewDrive feasibility assessment

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	DWD-FEA-001	Version	0.1
Status	Draft	Date	2026-10-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/dewdrive		
Source	docs/08-feasibility.md @ ca02742		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	First assessment (kit 1.8 pipeline), after the STEM check and DWD-CAL-001 v0.8

DewDrive feasibility assessment

DewDrive is feasible to build and test as a research prototype, but on paper it does not meet its 0.5 L per day target: the physics holds after the five STEM check fixes and every part can be made in a small workshop or bought, yet the design collects about 0.44 L per day (0.37 to 0.41 L under the sensitivities), on an isotherm built from bulk salt data and an estimated absorber absorptance. Method: STANDARDS section 23.

Table 1. Feasibility by aspect.

Aspect	Assessment	Rating	Risks and notes	Evidence
Design (physics)	Solar-regenerated sorption with a shaded condenser is within known science; 27 claims checked on option A, five detail failures fixed (absorber, tray floor, brine-wetted materials, condenser emissivity, anchors)	Medium	Yield 0.44 L per day against 0.5, PR-001 not met; 0.37 to 0.41 under the sensitivities (C9, C12, C13); recovery open for Amish	DWD-STM-001; DWD-CAL-001 v0.8
Materials	Plywood and PIR walls, aluminium sheet, PTFE coatings, twin-wall polycarbonate rated 130 °C, food-grade silica gel and $CaCl_2$	Medium	Coating quality and quote; water-resistant silica gel grade; polycarbonate grade against a 106 °C inner skin on a peak day; hot-material ratings (item 12)	DWD-DDR-005; DWD-DEC-001 items 1, 2, 9, 10
Manufacturing and assembly	Cut, fold, drill, rivet and bolt; 31 components and 19 assembly steps in the build plan; coatings bought in	High	Seal quality at the flaps, deck gasket and lid decides night bypass and day vapour loss	DWD-BLD-001; `cad/fidelity.json` (D3, 0 interferences)
Power and controls	10 W panel, 76.8 Wh LiFePO4 battery and an ESP32 logger with a fan timer and humidity cut-out; 23.6 Wh per day against 35 Wh	High	Firmware is a sketch until TRL 4; the cut-out needs a working RH sensor (fail-safe: fan off)	DWD-CAL-001 F1 to F3
Supply of parts	General suppliers for everything; the absorber sheet (R3 T6) and PTFE-coated glass-fibre mesh to confirm locally	Medium	Coating shop and mesh availability in the partner's region	BOM; DWD-DEC-001 items 9 to 12
Cost	$695 against a $520 value-engineering target	Medium	Over target by $175; water about $0.87 per litre over 5 years, parts only	DWD-CAL-001 J1, J2; DWD-MKT-001